

Sum of all divisors

Problem statement

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You are given an integer ' n '.

Function ' $\text{sumOfDivisors}(n)$ ' is defined as the sum of all divisors of ' n '.

Find the sum of ' $\text{sumOfDivisors}(i)$ ' for all ' i ' from 1 to ' n '.

Example:

Input: ' n ' = 5

Output: 21

Explanation:

We need to find the sum of ' $\text{sumOfDivisors}(i)$ ' for all ' i ' from 1 to 5.

$\text{sumOfDivisors}(1) = 1$

$\text{sumOfDivisors}(2) = 2 + 1 = 3$

$\text{sumOfDivisors}(3) = 3 + 1 = 4$

$\text{sumOfDivisors}(4) = 4 + 2 + 1 = 7$

$\text{sumOfDivisors}(5) = 5 + 1 = 6$

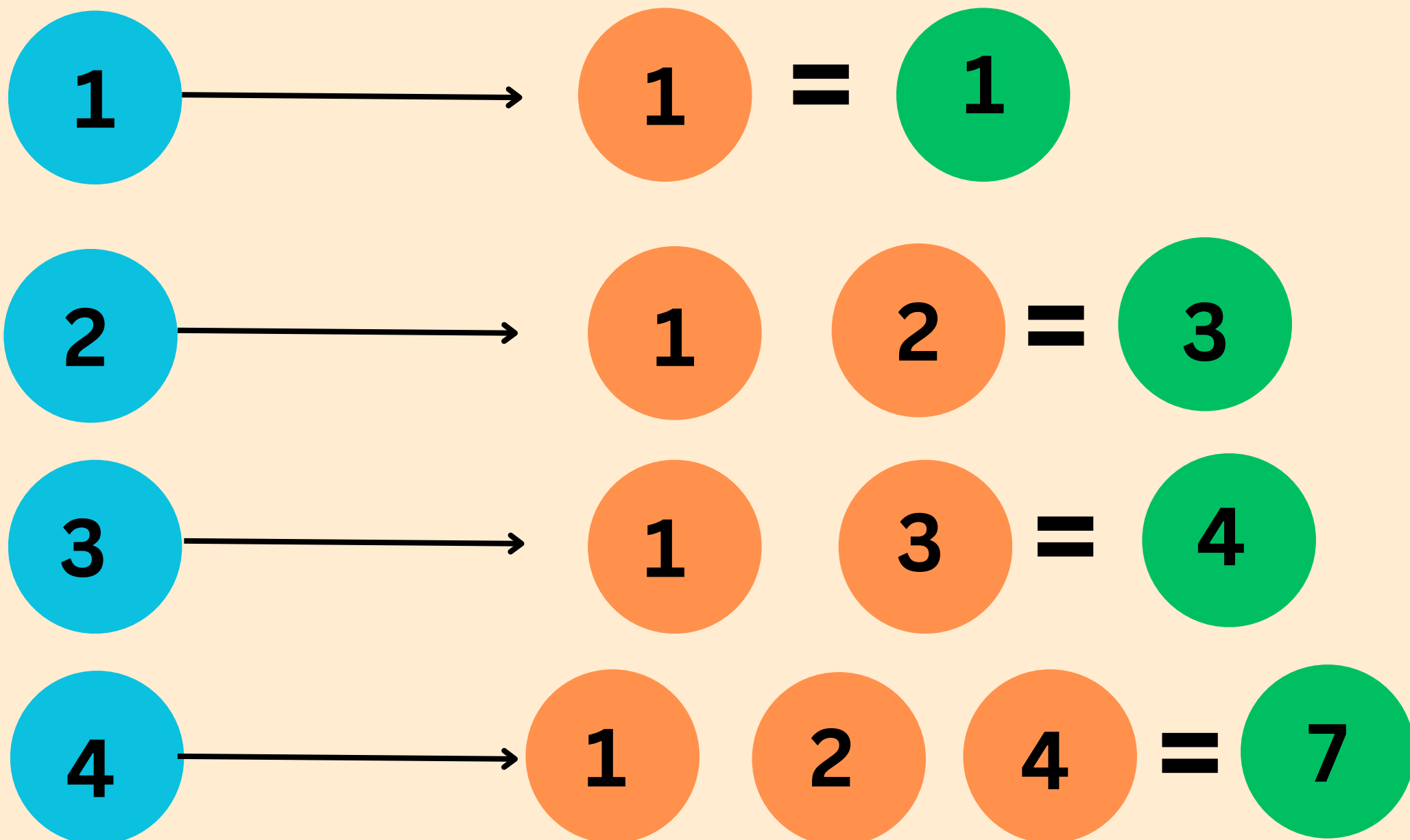
Therefore our answer is $\text{sumOfDivisors}(1) + \text{sumOfDivisors}(2) + \text{sumOfDivisors}(3) + \text{sumOfDivisors}(4) + \text{sumOfDivisors}(5) = 1 + 3 + 4 + 7 + 6 = 21$.

let's understand from examples:

$N = 4$

sum of all divisor of 1 to 4.

- write all divisor of 1 to 4.



$$1 + 3 + 4 + 7 = 15$$

Brute force :-

C++ (g++ 5.4) ▾

```
1  int sumOfAllDivisors(int n){
2      int sum = 0;
3      for(int i=1; i<=n; i++)
4      {
5          for(int j=1; j<=i; j++)
6          {
7              if(i % j == 0)
8              {
9                  sum += j;
10             }
11         }
12     }
13     return sum;
14 }
```

Time limit exceeded

A few seconds ago

Test cases

30/31

EXP

 38/40

Language

C++

Penalty

0%



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We think Common causes of Time Limit Exceeded :

- Infinite Loop
- Takes too much time than allowed constraints

Time complexity: $O(n*n)$

Space complexity: $O(1)$

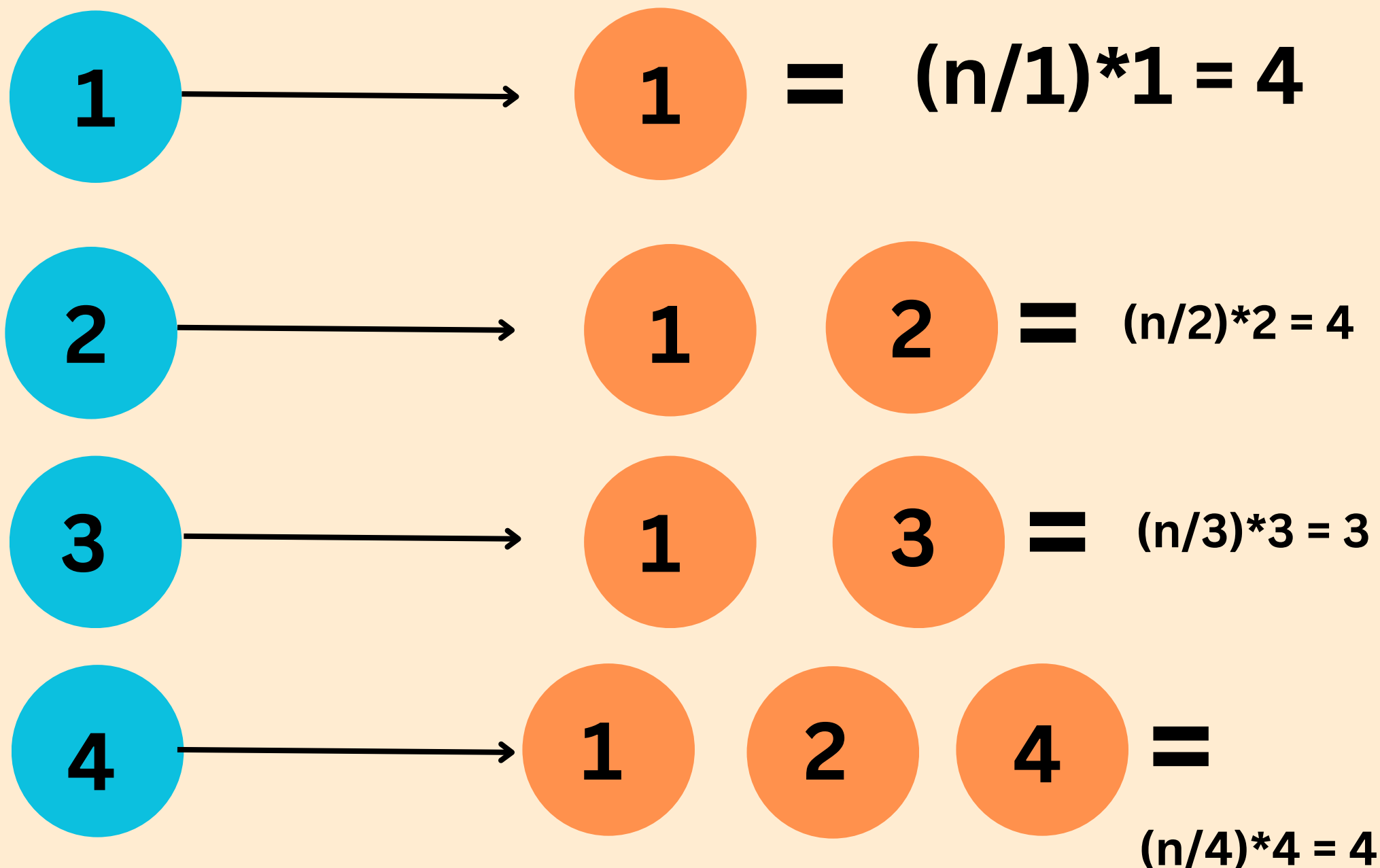
optimized approach:

let's understand from examples:

N = 4

sum of all divisor of 1 to 4.

- write all divisor of 1 to 4.



$$(N/i)*i$$

C++ (g++ 5.4) ▾

```
1 int sumOfAllDivisors(int n){
2     long long sum = 0;
3     for(int i=1; i<=n; i++)
4     {
5         sum += (n/i)*i;
6     }
7     return sum;
8 }
```

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time complexity: $O(n)$
space complexity: $O(1)$



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